Total No. of Questions : 8]

SEAT No. :

**P801**

[Total No. of Pages : 3

# [5870]-1119
# T.E. (Chemical)
# MASS TRANSFER - II
# (2019 Pattern) (Semester - II)

*Time : 2½ Hours]* *[Max. Marks : 70*

***Instructions to the candidates:***

1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figure to the right side indicate full marks.*
4) *Assume suitable data if necessary.*

***Q1)*** a) Define the following terms in solvent Extraction with their significance **[6]**

i) Selectivity

ii) Distribution coefficient

iii) Plait point

b) Explain the selection criteria of solvent used for Liquid-Liquid Extraction. **[6]**

c) A solution of Nicotine in water containing 1% nicotine is to be extracted with kerosene as a solvent at 293K. Water and kerosene are practically immiscible (Essentially insoluble).

Assume the equilibrium relationship to be $Y = 0.9\ X$

Where $Y$ = kg of nicotine / kg of kerosene

$X$ = kg of nicotine / kg of water

Determine % extraction of nicotine if 100 kg of the feed solution is extracted with 150 kg of solvent. **[6]**

OR

***Q2)*** a) Carbon Disulphide is used to extract iodine from its saturated solution in water in a single stage extraction. The distribution coefficient is given by $K = Y/X = 588.2$, where Y is gm of Iodine/ 1lit of $CS_2$ and X is gm of Iodine/ 1 lit of water. Calculate the concentration of Iodine in the aqueous phase if 1 litre of saturated aqueous solution containing 0.3 gm of Iodine / litre of water at 293K and contacted with 50ml of $CS_2$ by stirring. **[6]**

*P.T.O.*

b) Write a material balance for single stage solvent Extraction with immiscible solvents. **[6]**

c) Describe the principle, construction and mechanism of pulsed column for solvent Extraction with neat sketch. **[6]**

***Q3)*** a) Write the material balance for the continuous crosscurrent leaching assuming variable underflow and no insolubles in the overflow. **[6]**

b) Explain the constant and variable underflow in leaching operation. **[5]**

c) Discuss on the graphical representation of equilibrium characteristics of Leaching operation for constant underflow and no solid in overflow with diagram and proper notations. **[6]**

OR

***Q4)*** a) Oil is be extracted from meal by means of benzene using continuous countercurrent leaching unit. The unit treats 1000 kg of meal (on completely exhausted solids basis) per hour. The untreated meal contains 365 kg of oil and 30 kg of benzene. The solvent used contains 14 kg of oil and 590 kg of benzene. The exhausted solids contain 55 kg of oil and 451 kg of benzene. Find the number of stages required. The entrainment data is : **[10]**



| Kg of oil/kg of solution | 0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 |
|---|---|---|---|---|---|---|---|---|
| Kg of solution/kg of solid | 0.5 | 0.505 | 0.515 | 0.53 | 0.55 | 0.517 | 0.595 | 0.62 |

b) Enlist the equipment for leaching operation and explain any of them in detail. **[7]**

***Q5)*** a) A solution of washed raw cane sugar is colored by the presence of small amounts of impurities. The solution is to be decolorized by treatment with an adsorptive carbon in a contact filtration plant. The original solution has a color concentration of 9.6 measured on an arbitrary scale and it is desired to reduce colour of 0.96. Calculate the necessary dosage of the fresh carbon per 2000 kg solution for a single stage process. The data for an equilibrium isotherm is as follows: **[10]**

| Kg carbon/kg solution | 0 | 0.001 | 0.004 | 0.008 | 0.02 | 0.04 |
|---|---|---|---|---|---|---|
| Equilibrium colour | 9.6 | 8.6 | 6.3 | 4.3 | 1.7 | 0.7 |

b) Which parameters should affect the shape of Breakthrough curve in adsorption. **[8]**

OR

***Q6)*** a) Describe the Freundlich and Langmuir adsorption Isotherms in adsorption operation. **[6]**

b) Discuss on the principle, equilibria and rate of ion Exchange process in details. **[8]**

c) Discuss the applications of absorption operation. **[4]**

***Q7)*** a) Give the classification of crystallizers. Explain the construction and working of Swenson Walker Crystallizer with diagram. **[7]**

b) A Solution contains 500 Kg $Na_2CO_3$ and water has a concentration of 25% by wt. of salt. It is cooled from 335K to 285K in an agitated mild steel vessel. Weight of the vessel is 750 Kg. 2.0 % water is lost by evaporation. Crystals of $Na_2CO_3$. 10 $H_2O$ are formed. Calculate the yield of crystals and the heat to be removed? **[10]**

**Data:** Solubility At 285K: 8.9 Kg/ 100 Kg water.

Heat capacity of solution: 3.6 KJ / Kg K.

Heat Capacity of M.S: 0.5 KJ/ KgK.

Heat of Solution: 78.5 MJ / KMo1.

Latent heat of Vaporization: 2395 KJ/ Kg.

OR

***Q8)*** a) Define the terms in membrane processes; **[4]**

i) Rejection

ii) Permeability

iii) Membrane fouling

iv) Cake Resistance

b) What are different membrane modules? **[3]**

c) Write the material and energy balance of crystallizer. **[10]**

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